Smart Student Calculator

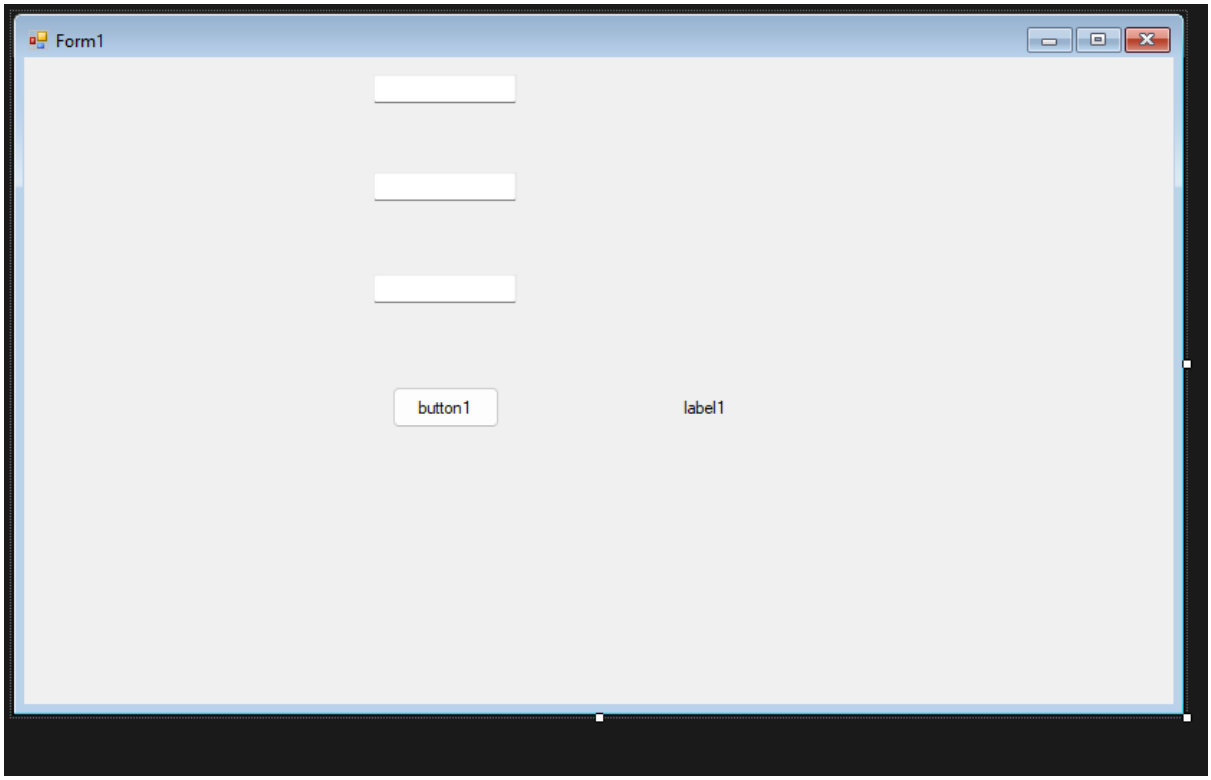
Code

```
using System;

0 references
class Program
{
    0 references
    static void Main(string[] args)
    {
        Console.WriteLine("Student Profile Processor ");

        string name = GetName();
        int age = GetAge();
        int marks = GetMarks();

        string readiness = CalculateReadiness(marks);

        DisplayProfile(name, age, marks, readiness);

        Console.ReadLine();
    }

    1 reference
    static string GetName()
    {
        Console.Write("Enter name: ");
        return Console.ReadLine();
    }

    1 reference
    static int GetAge()
    {
        int age;
        Console.Write("Enter age: ");

        while (!int.TryParse(Console.ReadLine(), out age))
        {
            Console.Write("Invalid age. Enter a number: ");
        }

        return age;
    }

    1 reference
    static int GetMarks()
    {
        int marks;
        Console.Write("Enter marks (0-100): ");

        while (!int.TryParse(Console.ReadLine(), out marks) || marks < 0 || marks > 100)
        {
            Console.Write("Invalid marks. Enter 0-100: ");
        }

        return marks;
    }

    // METHOD OVERLOADING (IMPORTANT FOR MARKS)
    1 reference
    static string CalculateReadiness(int marks)
    {
        if (marks >= 75)
            return "High";
        else if (marks >= 50)
            return "Medium";
        else
            return "Low";
    }

    0 references
    static string CalculateReadiness(int marks, int age)
    {
        if (marks >= 75 && age >= 18)
            return "High";
        else if (marks >= 50)
            return "Medium";
        else
            return "Low";
    }

    1 reference
    static void DisplayProfile(string name, int age, int marks, string readiness)
    {
        Console.WriteLine("\nStudent Profile");
        Console.WriteLine($"Name: {name}");
        Console.WriteLine($"Age: {age}");
        Console.WriteLine($"Marks: {marks}");
        Console.WriteLine($"Readiness: {readiness}");
    }
}
```

Program.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace StudentProfileForm
{
    0 references
    internal static class Program
    {
        /// <summary>
        /// The main entry point for the application.
        /// </summary>
        [STAThread]
        0 references
        static void Main()
        {
            Application.EnableVisualStyles();
            Application.SetCompatibleTextRenderingDefault(false);
            Application.Run(new Form1());
        }
    }
}
```

Form Design

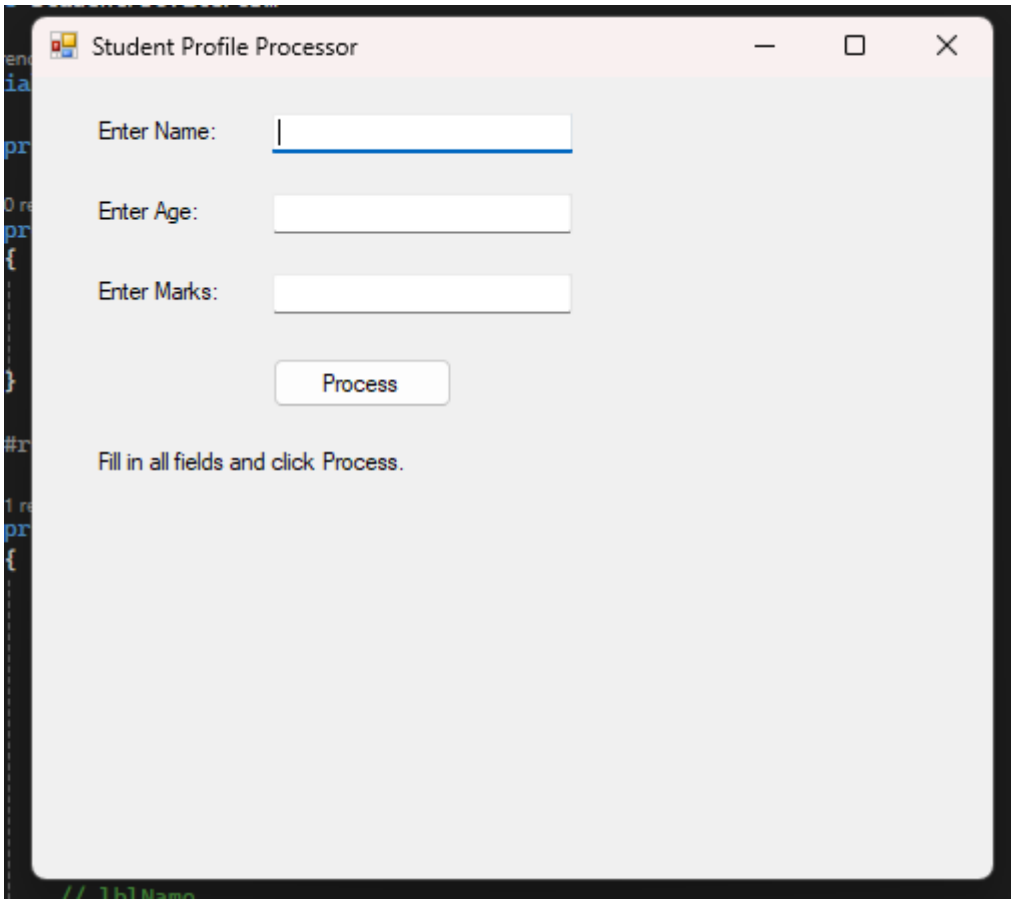


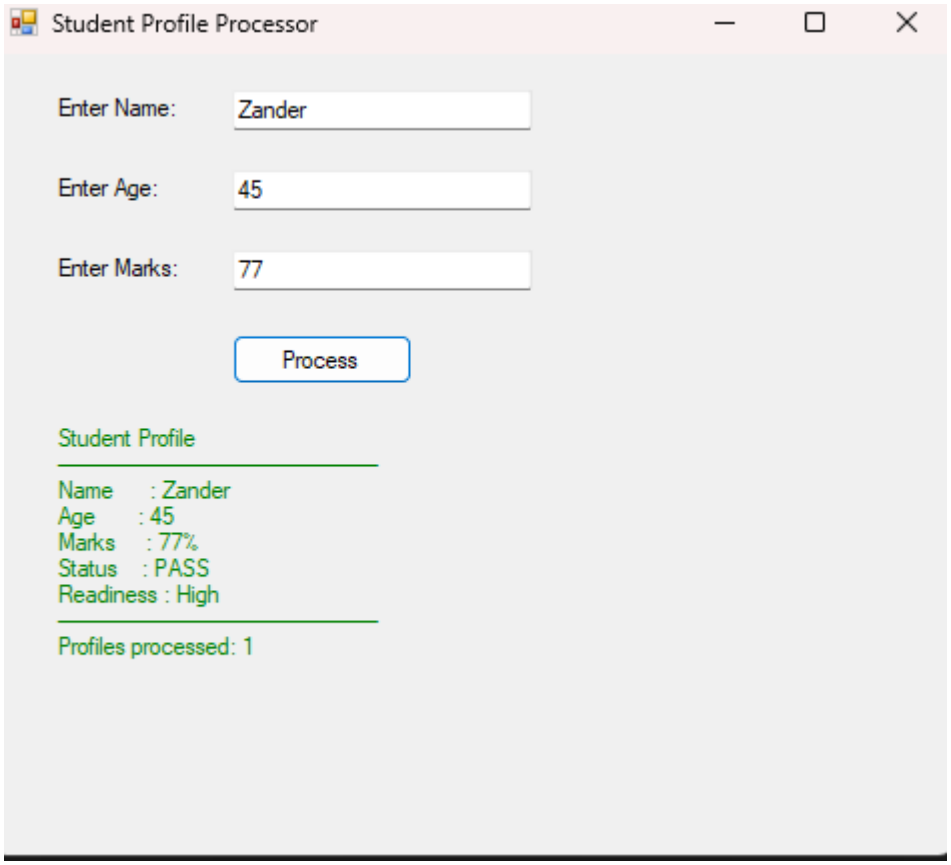
```
namespace StudentProfileForm
{
    3 references
    partial class Form1
    {
        /// <summary>
        /// Required designer variable.
        /// </summary>
        private System.ComponentModel.IContainer components = null;

        /// <summary>
        /// Clean up any resources being used.
        /// </summary>
        /// <param name="disposing">true if managed resources should be disposed; otherwise, false.</param>
        0 references
        protected override void Dispose(bool disposing)
        {
            if (disposing && (components != null))
            {
                components.Dispose();
            }
            base.Dispose(disposing);
        }

        Windows Form Designer generated code

        private System.Windows.Forms.TextBox txtName;
        private System.Windows.Forms.TextBox txtAge;
        private System.Windows.Forms.TextBox txtMarks;
        private System.Windows.Forms.Button btnProcess;
        private System.Windows.Forms.Label lblOutput;
    }
}
```

Output

```
Student Profile Processor
Enter name: Zander du Plessis
Enter age: 16
Enter marks (0-100): 70

Student Profile
Name: Zander du Plessis
Age: 16
Marks: 70
Readiness: Medium
```

Form


Student Profile Processor
Enter Name:
Enter Age:
Enter Marks:
Process
Fill in all fields and click Process.


Student Profile Processor
Enter Name:
Zander
Enter Age:
45
Enter Marks:
77
Process
Student Profile
Name : Zander
Age : 45
Marks : 77%
Status : PASS
Readiness : High
Profiles processed: 1